

CHAPTER 3: REQUIREMENTS AND ANALYSIS

3.1 Problem Definition

In today's digital era, event management has become an essential activity for educational institutions, corporate organizations, event management companies, and community groups. Many organizations still rely on manual methods such as paper-based registrations, spreadsheets, physical ticket distribution, and manual attendance records. These traditional approaches consume significant time, increase administrative workload, and are prone to errors such as duplicate registrations, data loss, and inefficient record management.

Manual event management also makes it difficult for users to obtain updated information regarding upcoming events, ticket availability, and booking confirmations. Organizers often face challenges in managing participant details, tracking payments, generating reports, and maintaining communication with attendees.

To overcome these challenges, there is a need for an automated web-based solution that provides a centralized platform for event creation, ticket booking, payment management, and participant record maintenance. The proposed **Smart Event Management and Ticket Booking System** addresses these issues by digitizing the complete event management process, improving operational efficiency, minimizing manual effort, and enhancing the overall user experience.

3.2 Requirements Specification

The system requirements are divided into functional and non-functional requirements.

3.2.1 Functional Requirements

The functional requirements describe the features that the system must provide.

- User Registration
 - User Login and Authentication
 - Profile Management
 - Event Creation by Administrator
 - Event Modification and Deletion
 - Event Search and Filtering
 - Ticket Booking
 - Booking History
 - Payment Management
 - Feedback Submission
 - Administrator Dashboard
 - User Management
 - Booking Management
 - Event Reports
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3.2.2 Non-Functional Requirements

The non-functional requirements describe the quality attributes of the system.

- User-friendly interface
 - Responsive web design
 - Fast response time
 - Secure user authentication
 - Data consistency
 - High system availability
 - Easy maintenance
 - Scalability
 - Cross-browser compatibility
 - Reliable database storage
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3.3 Planning and Scheduling

The project development follows a systematic software development life cycle (SDLC). The work is divided into multiple phases to ensure timely completion and effective management of the project.

Phase	Activity	Duration
Phase 1	Requirement Gathering	3 Days
Phase 2	System Analysis	2 Days
Phase 3	Database Design (ER Diagram)	3 Days
Phase 4	User Interface Design	4 Days
Phase 5	Frontend Development	8 Days
Phase 6	Backend Development	8 Days
Phase 7	Database Integration	4 Days
Phase 8	Testing and Debugging	4 Days
Phase 9	Documentation and Deployment	3 Days

PERT Sequence

Requirement Gathering



System Analysis



Database Design



UI Design



Frontend Development



Backend Development



Database Integration



Testing



Documentation & Deployment

3.4 Software and Hardware Requirements

3.4.1 Hardware Requirements

The following hardware configuration is sufficient for the development and execution of the proposed system.

Component	Specification
Processor	Intel Core i3 / Apple M1 or higher
RAM	Minimum 4 GB (8 GB Recommended)
Storage	20 GB Free Disk Space
Display	1366 × 768 Resolution or Higher
Internet	Broadband Internet Connection

3.4.2 Software Requirements

Software	Purpose
Windows 10/11 or macOS	Operating System
Visual Studio Code	Source Code Editor
Node.js	Backend Runtime Environment
Express.js	Backend Framework
React.js	Frontend Library
MongoDB Atlas	Cloud Database
Git	Version Control
GitHub	Source Code Repository
Google Chrome / Microsoft Edge	Web Browser
Postman	API Testing

Chapter Summary

This chapter discussed the existing problems associated with manual event management systems and identified the need for an automated web-based solution. It also defined the functional and non-functional requirements of the proposed system, outlined the project planning and scheduling using a phased development approach, and specified the software and hardware requirements necessary for the successful implementation of the Smart Event Management and Ticket Booking System.